SafeConnectKey

Validation Report - Secure Host and Computational Storage Communication

Profile	Run	Mode	Generated evidence
Recommended	20261025T164515Z-a718bdf7	full	Executed tests and raw CSV records

Executive summary

The local software prototype passed **18 of 18** required V01-V18 tests. Across 600 measured handshakes, median latency was **50.4439 ms** and p95 was **59.493845 ms**. AES-256-GCM, HMAC-SHA-256, and HKDF-SHA-256 are the only primary-profile cryptographic primitives.

Environment

Field	Measured value
CPU	Intel64 Family 6 Model 154 Stepping 4, GenuineIntel
RAM	15.73 GB
OS	Windows-11-10.0.26300-SP0
Python	3.13.13
Execution	local in-process FastAPI TestClient
Dependency hash	772e8cdd4712ca429c84b1c97d004f6b697a8b8f62f87188f782859b432b3aea

Measurement discipline

Raw records were written per operation before analysis. Sample payloads have exact documented byte sizes and SHA-256 digests. Targets are shown separately from measured values. No key, full token, or protected plaintext is included in this report.

V01-V18 functional and security results

ID	Expected	Observed	Status
V01	Session established and key fingerprints match.	passed	PASS
V02	Wrong AuthK is rejected; no session is created.	passed	PASS
V03	Wrong EnK fails authenticated decryption.	passed	PASS
V04	Modified AuthTd is rejected.	passed	PASS
V05	Modified SessTd is rejected by AES-GCM.	passed	PASS
V06	Expired challenge is rejected.	passed	PASS
V07	Consumed handshake is rejected as replay.	passed	PASS
V08	Substituted identity fails authentication.	passed	PASS
V09	Encrypted SHA-256 job returns the correct digest.	passed	PASS
V10	Modified job ciphertext is rejected without execution.	passed	PASS
V11	Repeated message sequence is rejected without duplicate execution.	passed	PASS
V12	Skipped sequence is rejected as out of order.	passed	PASS
V13	Expired session requires re-authentication.	passed	PASS
V14	Unknown operation returns a validated error.	passed	PASS
V15	Traversal outside csd_data is rejected.	passed	PASS
V16	Oversized request is rejected before processing.	passed	PASS
V17	Ten concurrent sessions remain isolated and complete.	passed	PASS
V18	Old session fails after restart and new handshake succeeds.	passed	PASS

Measured performance

Count	Median ms	p95 ms	p99 ms	Minimum ms	Maximum ms	Failures
600	50.4439	59.493845	63.041813	43.8485	113.8628	0

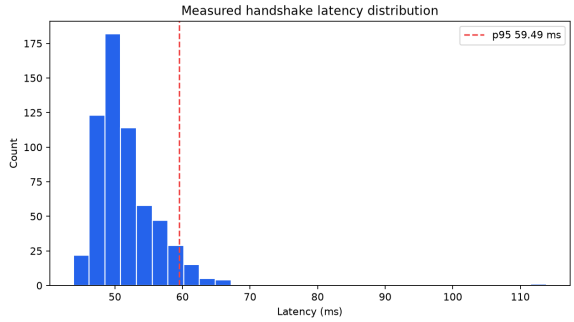
Payload cryptography

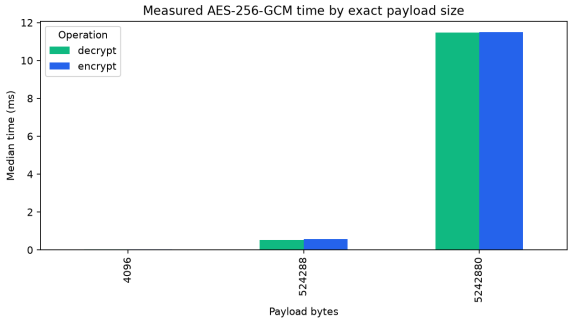
Exact bytes	Operation	Count	Median ms	p95 ms	Median MiB/s
4096	decrypt	600	0.028	0.033715	139.509
4096	encrypt	600	0.0398	0.07193	98.147
524288	decrypt	300	0.5281	0.75438	946.79
524288	encrypt	300	0.5722	0.88916	873.821
5242880	decrypt	90	11.4646	13.205625	436.125
5242880	encrypt	90	11.4995	13.36003	434.802

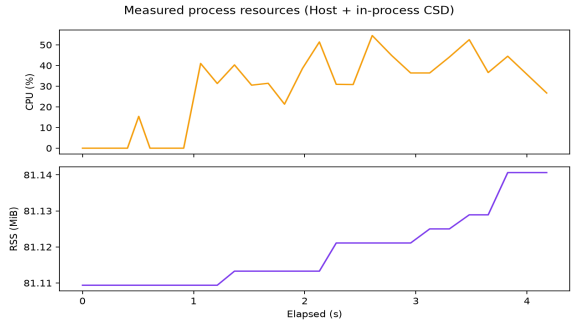
Near-data and concurrency

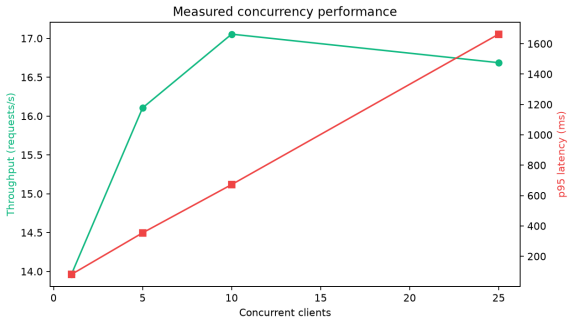
Metric	Measured value
Verified near-data jobs	1500 / 1500
Source bytes avoided	1733643600
10-client throughput	17.054 requests/s
10-client p95	672.187 ms
10-client error rate	0.0%
Peak combined RSS	81.141 MiB

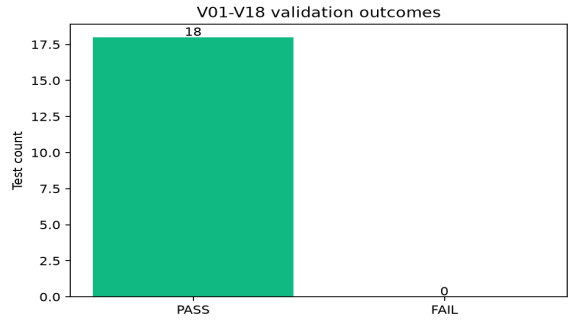
Measured charts











Acceptance targets versus measurements

Criterion	Target (not measured)	Measured	Outcome
V01-V18 pass rate	100%	100.0%	PASS
Handshake p95	< 100 ms	59.493845 ms	PASS
10-client error rate	< 1%	0.0%	PASS
Peak RSS	< 256 MiB	81.141 MiB*	PASS

Limitations

- Resource measurements include the Host and in-process FastAPI test harness, so they are not direct-process CSD-only measurements.

- No physical computational-storage hardware or resource-limited container was used.

- The narrated OBS video requires manual screen and microphone capture.

Benchmark mode was **full**. A plan-conformant full run uses the --full option: three rounds, 200 measured handshakes after 20 warm-ups per round, documented crypto/job iterations, and 60 seconds at each concurrency level.